Project1

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1. Introduction

LEGO sets are a global toy brand and a major collector's market, understanding what drives the price of a LEGO set has practical relevance for consumers, collectors, and manufacturers. The Brickset dataset offers a rich opportunity to explore how features related to price impact the market. By analyzing these variables and applying model selection techniques, the goal is to identify the most influential predictors of LEGO set price and construct a model that prioritizes not only predictive accuracy but also interpretability. This analysis will also allow us to examine larger patterns in LEGO set design and pricing across themes and release years.

2. Research Questions

The Brickset LEGO dataset provides detailed information about LEGO sets including price, piece count, themes, year released, and other characteristics. Because LEGO sets vary widely in complexity and price or market value, this dataset offers an opportunity to explore what factors drive the price of a set and whether these relationships can be modeled effectively.

Our primary research question is:

1. What factors most strongly influence the retails price of a LEGO set?

To support this, we also consider several smaller questions:

- Does piece count explain most of the variation in price?
- Does the year of release influence the price after controlling for all other variables?

To address these questions, multiple linear regression, best subsets selection, and stepwise regression to identify the most influential predictors and construct a final model balances simplicity and predictive accuracy.

3. Data Description

The dataset used in this analysis is from Brickset, primarily known as an online LEGO database that catalogs official LEGO sets released, along with details about minifigures, parts, colors, set inventories, and release information. The dataset was accessed through Github.

Each row in the dataset represents a single LEGO set and contains measurements recorded at the time of the set's original retail release. These included the sets' US retail price, physical characteristics, packaging dimensions, recommended age range, customer rating information. Because Brickset is a real-world catalog of LEGO products, the dataset consists of authentic observations rather than simulated data.

The full dataset contains 1,079 LEGO sets, spanning multiple themes and product families. The filename "2025.csv" includes LEGO sets that were released in 2025 or close to with some dates in 2024. Retail price is recorded in US dollars, weight is measured in grams, and packaging dimensions(height, width, depth) are recorded in millimeters. The variable pieces represents the number of building elements included in the set, and minifigs refers to the count of included minifigure characters. The dataset also includes several categorical fields, theme, themeGroup, and category, classifying the sets into product types of families.

```
variable_table <- data.frame(
  Variable = c(
    "US_retailPrice", "pieces", "minifigs", "weight",
    "height", "width", "depth", "agerange_min",
    "rating", "reviewCount", "theme", "themeGroup", "category"
  ),

  Role = c(
    "Response",
    rep("Quantitative Predictor", 9),
    rep("Categorical Predictor", 3)
  ),

  Data_Type = c(
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Categorical", "Categorical", "Categorical"
  ),

  Description = c(
    "Original U.S. retail price of the LEGO set in U.S. dollars",
    "Total number of LEGO pieces",
    "Number of minifigures included in the set",
```

```
    "Weight of the LEGO set",
    "Height of the LEGO set",
    "Width of the LEGO set",
    "Depth of the LEGO set",
    "Minimum recommended age for the set",
    "Average user rating for the set",
    "Number of user reviews",
    "Specific LEGO theme associated with the set",
    "Broader grouping of LEGO themes",
    "Category or type of LEGO set"
  )
)
variable_table
```

```
         Variable                   Role   Data_Type
1  US_retailPrice               Response     Numeric
2          pieces Quantitative Predictor     Numeric
3        minifigs Quantitative Predictor     Numeric
4          weight Quantitative Predictor     Numeric
5          height Quantitative Predictor     Numeric
6           width Quantitative Predictor     Numeric
7           depth Quantitative Predictor     Numeric
8    agerange_min Quantitative Predictor     Numeric
9          rating Quantitative Predictor     Numeric
10    reviewCount Quantitative Predictor     Numeric
11          theme  Categorical Predictor Categorical
12     themeGroup  Categorical Predictor Categorical
13       category  Categorical Predictor Categorical
                                                 Description
1  Original U.S. retail price of the LEGO set in U.S. dollars
2                                 Total number of LEGO pieces
3                  Number of minifigures included in the set
4                                       Weight of the LEGO set
5                                       Height of the LEGO set
6                                        Width of the LEGO set
7                                        Depth of the LEGO set
8                         Minimum recommended age for the set
9                             Average user rating for the set
10                                      Number of user reviews
11                Specific LEGO theme associated with the set
12                            Broader grouping of LEGO themes
13                               Category or type of LEGO set
```

4. Data Cleaning

The original dataset contains 1079 LEGO sets. Several variables weight and packaging dimensions have large amounts of missing data, meaning that requiring weight for all observations reduces the usable sample substantially. Other fields, height, width, depth, and reviewCount, also contain missing values.

If the dataset were to only included data points that included all data and contain no missing values, including categorical descriptors, the dataset would be reduced down to 61 complete observations which is 5.7% of the original dataset. Because such a small subset may not be representative of the full collection of LEGO sets, it is necessary to evaluate the consequences of this reduction.

```
#load data and read csv
df <- read.csv("2025.csv")

#create a table that counts all the missing data
variables <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

missing_table <- data.frame(
  Variable = variables,
  Missing = colSums(is.na(df[variables]))
)

missing_table
```

```
                    Variable Missing
US_retailPrice US_retailPrice     448
pieces               pieces     271
```

```
minifigs             minifigs     621
weight                 weight     893
height                 height     421
width                   width     421
depth                   depth     430
agerange_min     agerange_min     270
rating                 rating       0
reviewCount       reviewCount       0
theme                   theme       0
themeGroup         themeGroup       1
category             category       0
```

Variables with high coverage, such as retail price, piece count, and minifigure count will be used to compare retained and excluded sets. This will help determine whether the complete-case subset disproportionately contains sets from particular themes, sizes, and price ranges.

Zero values are also examined to see if it counts towards missingness or true measurements. A minifigure count of zero is a valid characteristic for certain sets, and a review count of zero may indicate that no reviews were submitted rather than missing data. These values were retained.

Because requiring weight excludes majority of the dataset, we adopt a staged analysis:

1. High-Coverage Model:

- Uses variables with minimal missingness(price, pieces, minifigs, dimensions). This model retains hundreds of observations and provides a broader view of pricing relationships.

2. Rich but Small Model:

- Uses the all the predictor variables within the complete case set with no missing values, but has limited coverage.

These models will be compared to evaluate how the sample size affect model selection and inference.

To prepare the dataset for analysis, we performed several data cleaning steps to ensure accuracy and consistency across all observations. This process involved identifying missing values and removing incomplete cases. The cleaned dataset, referred to as *data_complete*.

```
#create a new table that omits all the missing data
variables <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
```

```
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

data_complete <- df[complete.cases(df[variables]), ]
nrow(df)
```

```
[1] 1079
```

```
nrow(data_complete)
```

```
[1] 61
```

```
colSums(is.na(data_complete[variables]))
```

```
US_retailPrice         pieces       minifigs         weight         height
             0              0              0              0              0
         width          depth   agerange_min         rating    reviewCount
             0              0              0              0              0
         theme     themeGroup       category
             0              0              0
```

5. Exploratory Data Analysis

We begin by examining the distribution of key variables in the full dataset. This analysis will help determine the characteristics of variables and how they impact the model selection process. Correlation between variables is also an important feature to be analyzed, so the relationship between variables can be determined.

```
# full dataset summaries
summary(df$US_retailPrice)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   4.99   19.99   39.99   66.34   79.99  999.99     448
```

```
summary(df$pieces)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
    1.0    42.0   205.5   479.7   582.5  9023.0     271
```

```
summary(df$minifigs)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  1.000   1.000   2.000   3.275   4.000  40.000     621
```

```
summary(df$weight)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
 0.0070  0.1512  0.3255  0.6092  0.6843  3.9030     893
```

```
summary(df$height)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   5.20   14.70   19.10   22.69   26.20   55.50     421
```

```
summary(df$width)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   7.58   17.00   26.20   27.15   37.80   55.70     421
```

```
summary(df$depth)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  0.200   4.500   6.100   6.552   7.600  38.300     430
```

```
summary(df$rating)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   0.000   1.055   3.400   4.800
```

```r
summary(df$reviewCount)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.0000  0.0000  0.1835  0.0000  4.0000
```

```r
# complete case subset summaries from data_complete
summary(data_complete$US_retailPrice)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   9.99   19.99   39.99   58.92   79.99  299.99
```

```r
summary(data_complete$pieces)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   42.0   154.0   261.0   518.7   678.0  3266.0
```

```r
summary(data_complete$minifigs)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    2.00    3.00    3.23    4.00   10.00
```

```r
summary(data_complete$weight)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.1200  0.3260  0.5210  0.9087  1.2340  3.9030
```

```r
summary(data_complete$height)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  12.20   19.10   26.00   24.01   28.20   48.00
```

```r
summary(data_complete$width)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  14.10   20.50   28.20   32.22   45.60   55.50
```

```r
summary(data_complete$depth)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   4.60    6.10    6.20    7.33    8.80   13.20
```

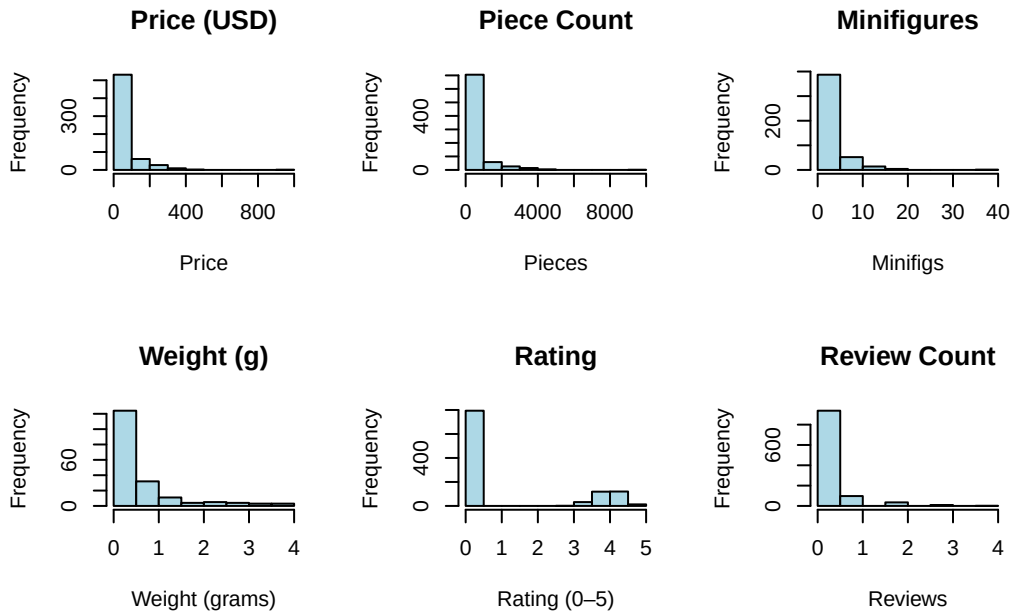
```r
summary(data_complete$rating)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   0.000   1.584   3.800   4.500
```

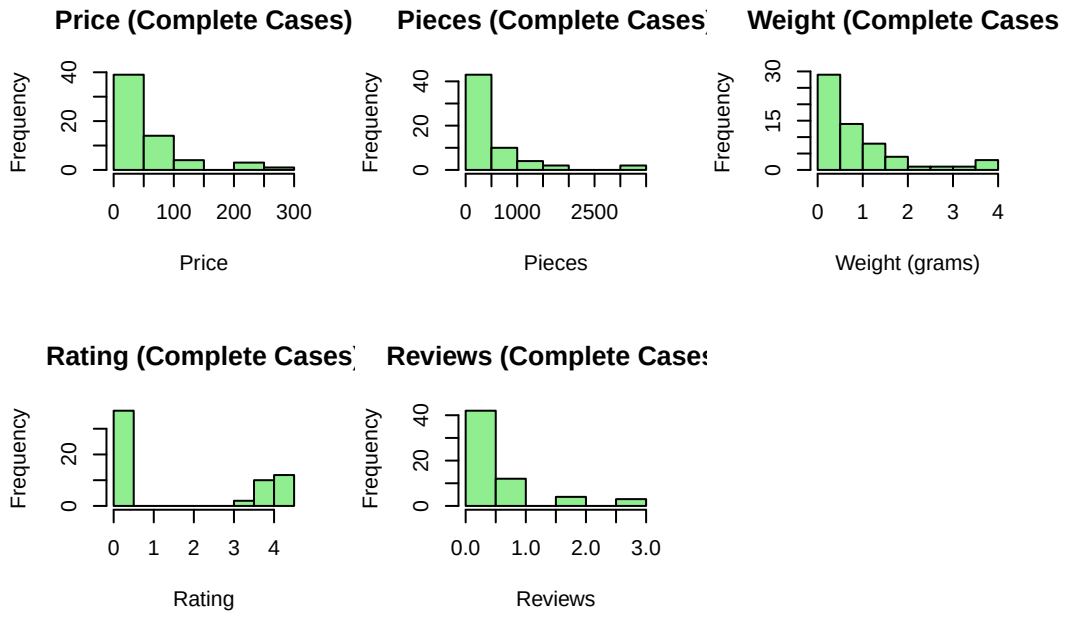
```r
summary(data_complete$reviewCount)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.0000  0.0000  0.4754  1.0000  3.0000
```

```r
#Histograms for full dataset vs the cleaned dataset
par(mfrow = c(2, 3))
hist(df$US_retailPrice, main = "Price (USD)", xlab = "Price", col = "lightblue")
hist(df$pieces, main = "Piece Count", xlab = "Pieces", col = "lightblue")
hist(df$minifigs, main = "Minifigures", xlab = "Minifigs", col = "lightblue")
hist(df$weight, main = "Weight (g)", xlab = "Weight (grams)", col = "lightblue")
hist(df$rating, main = "Rating", xlab = "Rating (0-5)", col = "lightblue")
hist(df$reviewCount, main = "Review Count", xlab = "Reviews", col = "lightblue")
```




```
# Histograms (61-case subset)
par(mfrow = c(2, 3))
hist(data_complete$US_retailPrice, main = "Price (Complete Cases)", xlab = "Price", col = "li
hist(data_complete$pieces, main = "Pieces (Complete Cases)", xlab = "Pieces", col = "lightgre
hist(data_complete$weight, main = "Weight (Complete Cases)", xlab = "Weight (grams)", col = '
hist(data_complete$rating, main = "Rating (Complete Cases)", xlab = "Rating", col = "lightgre
hist(data_complete$reviewCount, main = "Reviews (Complete Cases)", xlab = "Reviews", col = "l
```



Distribution in Full Dataset

Retail price shows right-skewness where most LEGO sets are clustered at lower prices and a long right tail where LEGO set prices increases and occurs less. The median price is $39.99, but the means rises to $66.34, showing that there is a small number of expensive sets that pull the average price upward. Prices range from $4.99 to $999.99 with 448 sets missing price.

Price count has similar behavior. The median set contains about 205.5 pieces, but the mean is about 479.7 pieces, showing a long right tail. There are clusters of LEGO sets with a smaller amount of pieces, and trails off as the amount of pieces increase. Piece count ranges from 1 to 9,023, with 271 missing values.

Minifigure counts also vary widely. Although the median is 2 minifigures, the mean is 3.275, and counts range from 1 to 40. 621 sets have missing minifigure data, and many sets include zero minifigures.

Weight has the highest missingness as 893 sets lack weight data, leaving only 186 sets with recorded weights. This extremely high missingness explains why complete-case subset shrinks so dramatically.

Packaging dimensions (height, width, depth) also show substantial missingness (421–430 missing values each). Among observed values, height ranges from 5.2 mm to 55.5 mm, width from 7.58 mm to 55.7 mm, and depth from 0.2 mm to 38.3 mm. These variables are right-skewed and strongly correlated with piece count and weight.

Ratings and review counts are heavily zero-inflated. The median rating is 0, and the median review count is 0, indicating that many sets have no user feedback recorded. Ratings range from 0 to 4.8, and review counts from 0 to 4.

Distribution within the Complete-Case Subset

The complete-case subset differs noticeably from the full dataset. Price remains right-skewed, but the range is narrower: \$9.99 to \$299.99, with a mean of \$58.92. This suggests that the complete cases under-represent very high-priced sets.

Piece counts also show reduced range, 42 to 3,266 pieces, compared to the full dataset's maximum of 9,023. The median piece count is 261, slightly higher than the full dataset median (205.5), indicating that the complete-case subset may over-represent medium-sized sets.

Minifigure counts are more concentrated: median 3, mean 3.23, and maximum 10, compared to 40 in the full dataset. This again suggests that the complete-case subset excludes many large or specialized sets.

Weight is fully observed in this subset by construction, with a median of 0.521 kg and mean of 0.9087 kg, substantially higher than the full dataset's median (0.3255 kg). This indicates that sets with recorded weight tend to be larger and heavier.

Dimensions also show higher averages in the complete-case subset. For example, median height 26 mm, median width 28.2 mm. This reinforces the idea that complete cases disproportionately include larger sets.

Ratings and review counts have higher means in the complete-case subset (mean rating 1.584, mean review count 0.475), suggesting that sets with complete metadata may be more popular or more frequently reviewed.

Key Takeaways

- Strong right-skewness in price, pieces, weight, and dimensions
- Zeros in ratings and review counts are to be interpreted as "no reviews" rather than missing data
- Weight and dimensions have a high amount of missing data, which is why the complete-case subset only leaves 61 LEGO sets to be observed
- Based on the summaries, the complete-case subset is not representative of the data because it contains heavier, larger sets, it excludes many high priced and high piece count LEGO sets, and it overerpresents sets with more complete data and reviews.

Correlation Analysis

```r
# quantitative variables

quantitative_vars <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount"
)

# Select quantitative variables
correlation_vars <- data_complete[quantitative_vars]

# Calculate the correlation matrix
correlation_matrix <- cor(
  correlation_vars,
  use = "complete.obs"
)

correlation_matrix
```

```
               US_retailPrice    pieces  minifigs    weight    height     width
US_retailPrice      1.0000000 0.9106009 0.7786630 0.9671660 0.8191912 0.7984531
pieces              0.9106009 1.0000000 0.7555396 0.8658698 0.6260305 0.6528074
minifigs            0.7786630 0.7555396 1.0000000 0.7486760 0.6631737 0.6829421
weight              0.9671660 0.8658698 0.7486760 1.0000000 0.8464519 0.8272257
height              0.8191912 0.6260305 0.6631737 0.8464519 1.0000000 0.8323138
width               0.7984531 0.6528074 0.6829421 0.8272257 0.8323138 1.0000000
depth               0.7667257 0.6624670 0.5304681 0.8274525 0.7266921 0.7192446
agerange_min        0.5728106 0.7147580 0.3859962 0.4524908 0.3079079 0.3201698
rating              0.2523957 0.3476234 0.2676433 0.2636961 0.1064871 0.1294318
reviewCount         0.2599065 0.3492340 0.2701237 0.2828723 0.1505186 0.1648387
                    depth agerange_min    rating reviewCount
US_retailPrice 0.7667257    0.5728106 0.2523957   0.2599065
pieces         0.6624670    0.7147580 0.3476234   0.3492340
minifigs       0.5304681    0.3859962 0.2676433   0.2701237
weight         0.8274525    0.4524908 0.2636961   0.2828723
```

```
height          0.7266921    0.3079079 0.1064871   0.1505186
width           0.7192446    0.3201698 0.1294318   0.1648387
depth           1.0000000    0.4066463 0.1718117   0.2724942
agerange_min    0.4066463    1.0000000 0.3509209   0.3342968
rating          0.1718117    0.3509209 1.0000000   0.5944375
reviewCount     0.2724942    0.3342968 0.5944375   1.0000000
```

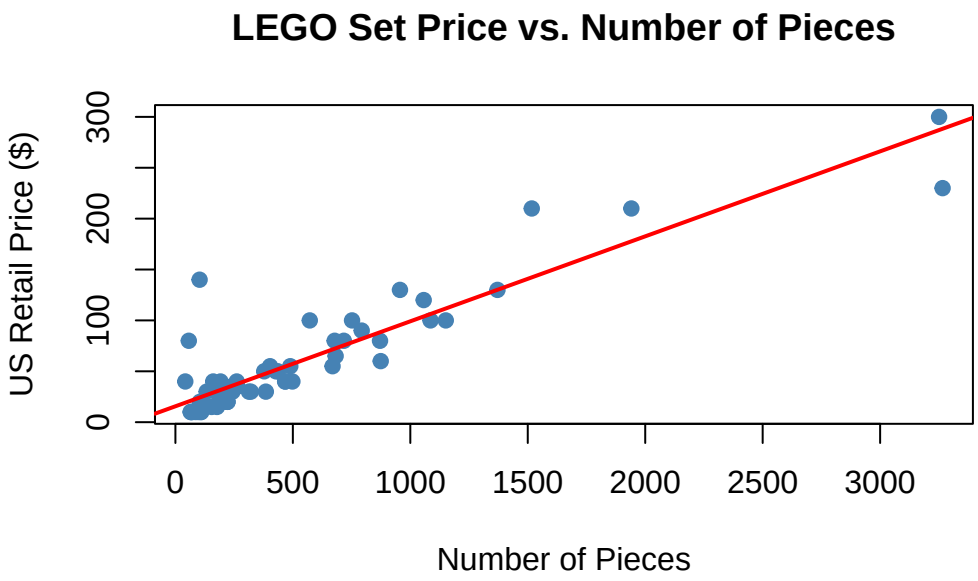
The correlation matrix shows strong positive relationships between LEGO set retail price and several measures of set size and complexity. The strongest correlation is between US retail price and weight (r = 0.967), followed by the number of pieces (r = 0.911). This suggests that heavier sets and sets containing more pieces tend to have higher retail prices. Price also has strong positive correlations with height (r = 0.819), width (r = 0.798), depth (r = 0.767), and the number of minifigures (r = 0.779). These results suggest that the physical size and contents of a LEGO set are closely associated with its retail price. The minimum recommended age also has a moderate positive correlation with price (r = 0.573), indicating that sets designed for older builders may tend to be more expensive. In contrast, user ratings (r = 0.252) and review counts (r = 0.260) have relatively weak correlations with price, suggesting that customer ratings and the number of reviews are less strongly associated with retail price than the physical characteristics of a set.

The matrix also reveals strong correlations among several predictor variables. For example, piece count and weight have a correlation of 0.866, while weight is also strongly correlated with height (r = 0.846), width (r = 0.827), and depth (r = 0.827). These relationships suggest that these variables could create multicollinearity in a multiple linear regression model. This means that including all these predictors simultaneously may make it harder to distinguish their individual contributions to price. Additionally, ratings and review counts have a moderate positive correlation (r = 0.594), suggesting that sets with more reviews may also tend to have higher ratings. Overall, the correlation matrix suggests that weight and piece count are promising predictors of retail price, although further regression analysis and model selection will be necessary to determine which variables provide the most useful independent explanatory power.

```r
#Scatter plots of key relationships

plot(
  US_retailPrice ~ pieces,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Pieces",
  xlab = "Number of Pieces",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "steelblue"
)
```

```r
# Add a fitted linear regression line
abline(
  lm(US_retailPrice ~ pieces, data = data_complete),
  col = "red",
  lwd = 2
)
```



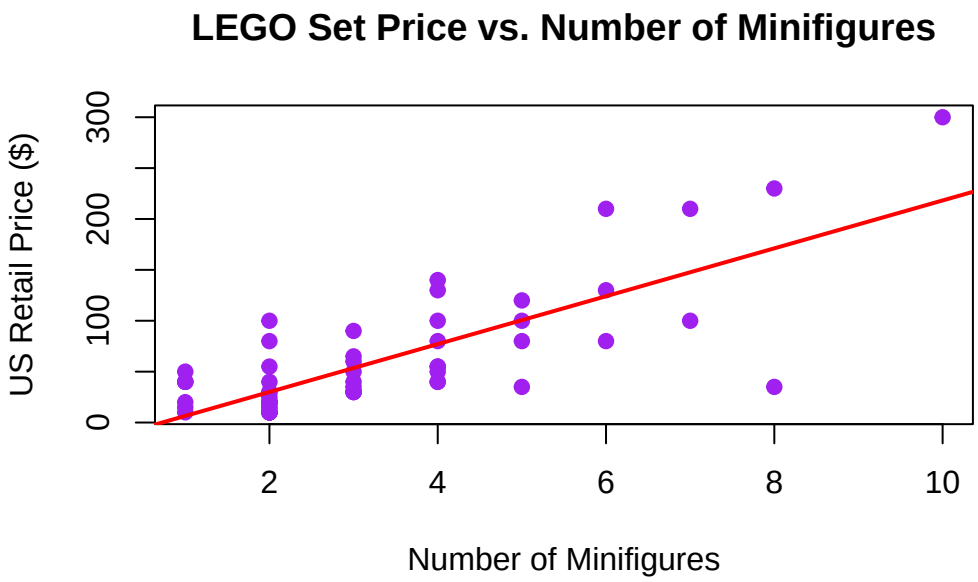

```r
plot(
  US_retailPrice ~ weight,
  data = data_complete,
  main = "LEGO Set Price vs. Weight",
  xlab = "Weight",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "darkgreen"
)

abline(
  lm(US_retailPrice ~ weight, data = data_complete),
  col = "red",
  lwd = 2
)
```




```
plot(
  US_retailPrice ~ minifigs,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Minifigures",
  xlab = "Number of Minifigures",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "purple"
)

abline(lm(US_retailPrice ~ minifigs, data = data_complete),col = "red",lwd = 2)
```



The scatterplots reveal positive relationships between LEGO set retail price and the number of pieces and weight. The strongest visual relationship appears in the plot of retail price versus weight, where price generally increases as the weight of a set increases. This is consistent with the correlation coefficient of 0.967, indicating a very strong positive linear association. The plot of retail price versus number of pieces also shows a clear upward trend, consistent with its correlation of 0.911. Sets with more pieces generally have higher retail prices, although there is some variation among sets with similar piece counts. Several observations lie farther from the regression line, suggesting that some sets are considerably more expensive or less expensive than their piece counts alone might predict. These observations may reflect differences in set design, theme, or other characteristics that influence pricing. The plots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

The relationship between retail price and the number of minifigures is also positive, but the pattern is less consistent than those observed for weight and piece count. Although sets with more minifigures tend to have higher prices, there is considerable variation in price among sets containing the same number of minifigures. For example, sets containing six minifigures have substantially different retail prices, suggesting that minifigure count alone does not fully explain a set's price. Overall, the scatterplots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

6. Best Subsets Regression

Best Subsets Regression is a model selecting method that takes the response variable, in this case US_retailPrice, and the predictors variables within the dataset and yields the possible models and their statistics. This method evaluates every possible combination of predictors and identifies the best model for each model size using adjusted R-squared, BIC and Cp. Adjusted R-squared tells us how well the predictors explain the variation in the response variable and the higher the adjusted R-squared the better. BIC tells us how simple and interpretable a model is; the lower the BIC the better the model. Cp tells how well a model balances fit and complexity; the lower the Cp the better the model.

Stage 1 - Large sample

Because several predictors in the full dataset have substantial missingness, especially weight, rating, and reviewCount, we pay attention to variables with relatively high coverage: pieces, minifigs, height, width, depth, and agerange_min. These variables allow us to retain a much larger portion of the dataset while still capturing key physical characteristics of LEGO sets.

```
# load the package
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.5.3
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```r
# counts the amount of observations(rows) in Stage 1
df_stage1 <- df %>% drop_na(US_retailPrice, pieces, minifigs, height, width, depth, agerange_

nrow(df_stage1)
```

```
[1] 289
```

After removing rows with missing values in these fields, the Stage 1 dataset contains 289 observations, more than the 61 complete cases available when weight and rating are required. This larger sample provides a more representative view of the overall LEGO product catalog and serves as the foundation for the high-coverage model selection.

```r
 # load package for best subsets regression
library(leaps)
```

```
Warning: package 'leaps' was built under R version 4.5.3
```

```r
# run best subsets
best_subsets <- regsubsets(US_retailPrice ~ pieces + minifigs +
                             height + depth + width + agerange_min,
                           data = df_stage1)


# display summary of the best regression models
best_summary_model <- summary(best_subsets)
best_summary_model
```

```
Subset selection object
Call: regsubsets.formula(US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min, data = df_stage1)
6 Variables  (and intercept)
             Forced in Forced out
pieces           FALSE      FALSE
minifigs         FALSE      FALSE
height           FALSE      FALSE
depth            FALSE      FALSE
width            FALSE      FALSE
agerange_min     FALSE      FALSE
1 subsets of each size up to 6
Selection Algorithm: exhaustive
```

```
         pieces minifigs height depth width agerange_min
1  ( 1 ) "*"    " "      " "    " "   " "   " "
2  ( 1 ) "*"    " "      " "    "*"   " "   " "
3  ( 1 ) "*"    "*"      " "    "*"   " "   " "
4  ( 1 ) "*"    "*"      "*"    "*"   " "   " "
5  ( 1 ) "*"    "*"      "*"    "*"   " "   "*"
6  ( 1 ) "*"    "*"      "*"    "*"   "*"   "*"
```

```
best_summary_model$adjr2
```

```
[1] 0.9263590 0.9414897 0.9521875 0.9568381 0.9566949 0.9565445
```

```
best_summary_model$cp
```

```
[1] 201.358340 102.081950  32.574978   3.080734   5.019948   7.000000
```

```
best_summary_model$bic
```

```
[1] -743.5443 -805.3563 -859.0556 -883.9784 -878.3743 -872.7283
```

```
#which tells which predictors are included in each model
best_summary_model$which
```

```
  (Intercept) pieces minifigs height depth width agerange_min
1        TRUE   TRUE    FALSE  FALSE FALSE FALSE        FALSE
2        TRUE   TRUE    FALSE  FALSE  TRUE FALSE        FALSE
3        TRUE   TRUE     TRUE  FALSE  TRUE FALSE        FALSE
4        TRUE   TRUE     TRUE   TRUE  TRUE FALSE        FALSE
5        TRUE   TRUE     TRUE   TRUE  TRUE FALSE         TRUE
6        TRUE   TRUE     TRUE   TRUE  TRUE  TRUE         TRUE
```

Applying the best-subsets regression to this 289 subset using price as the response variable, the strongest predictors were pieces, minifigs, height, and depth. The four-predictor model with these 4 predictors is the best model. This model achieves the highest adjusted R-squared (0.9568), indicating the strongest explanatory power among the other models. It also has the lowest Cp(3.08), which is close to the ideal value of p(predictors) + 1 = 5, and the lowest BIC(883.98), showing the best balance between fit and model complexity. Although, the five and six predictor models have adjusted R-squared values that are close to the four-predictor model, their Cp and BIC values are higher, indicating more complexity with little improvement. Therefore, the four-predictor model, model 4, is the preferred Stage 1 model.

Stage 2 - Complete Case

Stage 2 uses a wider range of the predictor variables but only uses LEGO sets where there is no missing values. This subset is considered rich but small as it only contains 61 observations. In this subset we are including weight because it has strong explanatory value and rating and reviewCount may matter, but because these variables have huge missingness, they must be evaluated separately from Stage 1.

```
df_stage2 <- df %>% drop_na(US_retailPrice, pieces, minifigs, height, width, depth, agerange_

nrow(df_stage2)
```

```
[1] 61
```

```
 # load package for best subsets regression
library(leaps)

# run best subsets
best_subsets <- regsubsets(US_retailPrice ~ pieces + minifigs +
                             height + depth + width + agerange_min +
                             weight + rating + reviewCount,
                           data = df_stage2)


# display summary of the best regression models
best_summary_model <- summary(best_subsets)
best_summary_model
```

```
Subset selection object
Call: regsubsets.formula(US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min + weight + rating + reviewCount,
    data = df_stage2)
9 Variables  (and intercept)
             Forced in Forced out
pieces           FALSE      FALSE
minifigs         FALSE      FALSE
height           FALSE      FALSE
depth            FALSE      FALSE
width            FALSE      FALSE
agerange_min     FALSE      FALSE
weight           FALSE      FALSE
```

```
rating           FALSE      FALSE
reviewCount      FALSE      FALSE
1 subsets of each size up to 8
Selection Algorithm: exhaustive
         pieces minifigs height depth width agerange_min weight rating
1  ( 1 ) " "    " "      " "    " "   " "   " "          "*"    " "
2  ( 1 ) " "    " "      " "    " "   " "   "*"          "*"    " "
3  ( 1 ) " "    "*"      " "    " "   " "   "*"          "*"    " "
4  ( 1 ) " "    "*"      " "    " "   " "   "*"          "*"    " "
5  ( 1 ) " "    "*"      " "    "*"   " "   "*"          "*"    "*"
6  ( 1 ) " "    "*"      " "    "*"   " "   "*"          "*"    "*"
7  ( 1 ) "*"    "*"      " "    "*"   " "   "*"          "*"    "*"
8  ( 1 ) "*"    "*"      "*"    "*"   " "   "*"          "*"    "*"
         reviewCount
1  ( 1 ) " "
2  ( 1 ) " "
3  ( 1 ) " "
4  ( 1 ) "*"
5  ( 1 ) " "
6  ( 1 ) "*"
7  ( 1 ) "*"
8  ( 1 ) "*"
```

```
best_summary_model$adjr2
```

```
[1] 0.9343154 0.9569526 0.9614188 0.9643101 0.9667275 0.9669445 0.9666173
[8] 0.9666360
```

```
best_summary_model$cp
```

```
[1] 56.953050 18.415108 11.663737  7.768334  4.809390  5.486479  7.024448
[8]  8.014201
```

```
best_summary_model$bic
```

```
[1] -158.8998 -181.6080 -185.2398 -186.9604 -188.2270 -185.6345 -182.0630
[8] -179.1483
```

```
#which tells which predictors are included in each model
best_summary_model$which
```

```
  (Intercept) pieces minifigs height depth width agerange_min weight rating
1        TRUE  FALSE    FALSE  FALSE FALSE FALSE        FALSE   TRUE  FALSE
2        TRUE  FALSE    FALSE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
3        TRUE  FALSE     TRUE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
4        TRUE  FALSE     TRUE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
5        TRUE  FALSE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
6        TRUE  FALSE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
7        TRUE   TRUE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
8        TRUE   TRUE     TRUE   TRUE  TRUE FALSE         TRUE   TRUE   TRUE
  reviewCount
1       FALSE
2       FALSE
3       FALSE
4        TRUE
5       FALSE
6        TRUE
7        TRUE
8        TRUE
```

Based on the best subsets regression for stage 2, the strongest predictors are weight, agerange, and minifigs. On the lower side of predictor impact is depth, rating, and reivewCount. Model 6 has the highest R-squared value (0.9669), indicating that it has the strongest explanatory prowess with Model 5, 7, and 8 all being around ~ 0.966. The model that balances fit and complexity the best is Model 5 with a Cp of 4.81, and it also has the best BIC of -188.23. Although Model 6 has the best adjusted R-squared, Model 5 only differs from Model 6 by 0.00217. The difference between the models is extremely small, and adding another predictor adds more complexity. Therefore, Model 5 keeps simplicity without losing key explanatory power.

Stage 1 vs Stage 2 Comparison

The staged model-selection strategy was necessary because the full dataset contains substantial missingness. These variables appear to have strong explanatory value, but requiring them reduces the usable sample from 1,079 sets to only 61 complete cases. By separating the analysis into two stages, we can evaluate how model conclusions change when richer predictors are included at the cost of dramatically reduced sample coverage.

Stage 1 uses predictors with relatively high coverage such as, pieces, minifigs, height, width, depth, and agerange_min, retaining 289 observations. This larger sample is more representative

of the overall LEGO catalog and provides a stable foundation for model selection. Stage 2 requires weight, rating, and reviewCount, which reduces the dataset to 61 complete cases. This smaller subset includes larger, heavier sets with more complete data, introducing potential selection bias.

Predictor Strength Predictor strength differs between stages. In Stage 1, pieces is the strongest predictor, followed by depth, minifigs, height, and agerange_min. These variables reflect physical size and complexity, which are widely available and strongly associated with price.

In Stage 2, weight becomes the strongest predictor, followed by agerange_min, minifigs, depth, and rating. Weight enters the model first and remains included in every larger model, indicating that it captures information about set size and complexity that is not fully represented by piece count or dimensions. Rating and reviewCount enter only in larger models, suggesting weaker adjusted associations.

Model Selection Stage 1 selects a four-predictor model (pieces, minifigs, height, depth) as the best model. It has the highest adjusted R^2, lowest Cp, and lowest BIC, indicating strong explanatory power with minimal complexity. Although five- and six-predictor models have similar adjusted R^2 values, their Cp and BIC values are higher, indicating more complexity.

Stage 2 selects a five-predictor model (weight, agerange_min, minifigs, depth, rating). Although the six-predictor model has the highest adjusted R^2, the difference is extremely small, and the five-predictor model has both the lowest Cp and lowest BIC, making it the preferred model.

Consequences of Missing Data The comparison highlights how missing values shapes model conclusions. Weight is a dominant predictor in Stage 2 but is missing for 893 sets in the full dataset. Ratings and review counts may reflect customer feedback collected after release. Stage 1 avoids these issues by focusing on predictors available for most sets, while Stage 2 explores the richer but selective subset.

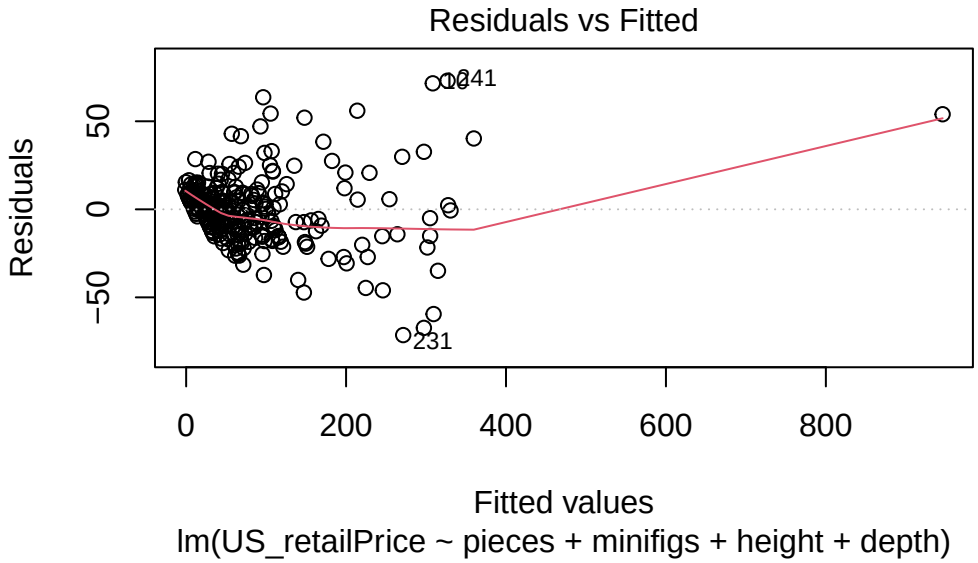
Model Interpretation Stage 1 provides a broad model based on widely available physical characteristics. Stage 2 provides a richer model that incorporates weight and rating but is based on a much smaller and potentially selective sample. The staged approach allows us to compare these models directly and understand how predictor availability and sample coverage influence model selection. Stage 1 supports general pricing patterns across the dataset, while Stage 2 describes associations within a narrow subset of sets.

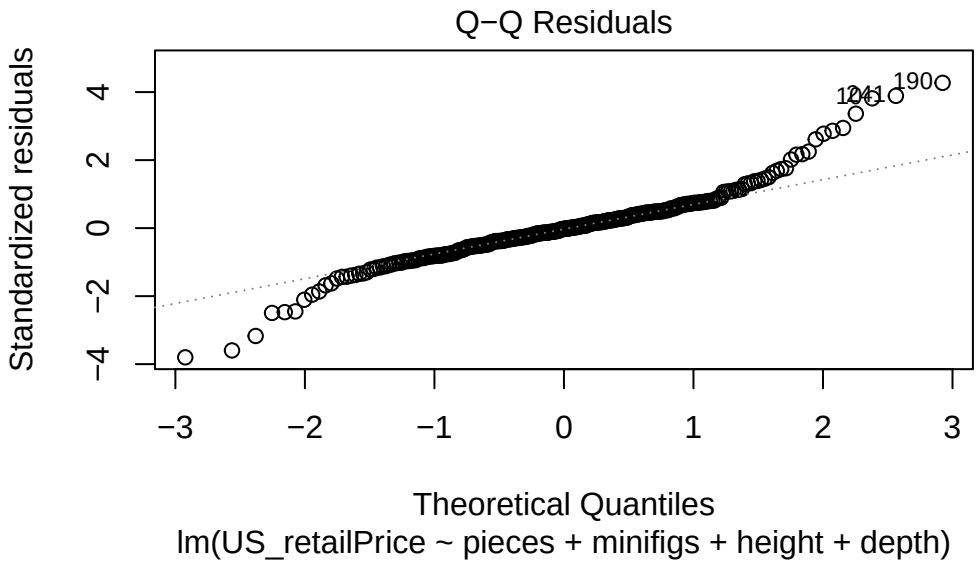
7. Model Diagnostics

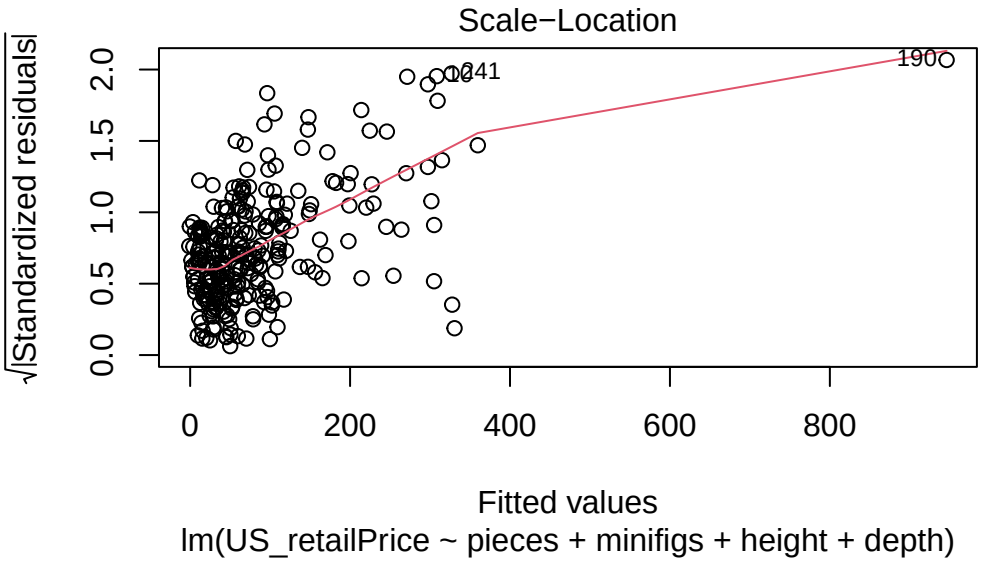
The diagnostic plots, residuals vs fitted, scale-location, q-q plot, cooks distance, will show whether the best subset model is actually valid.

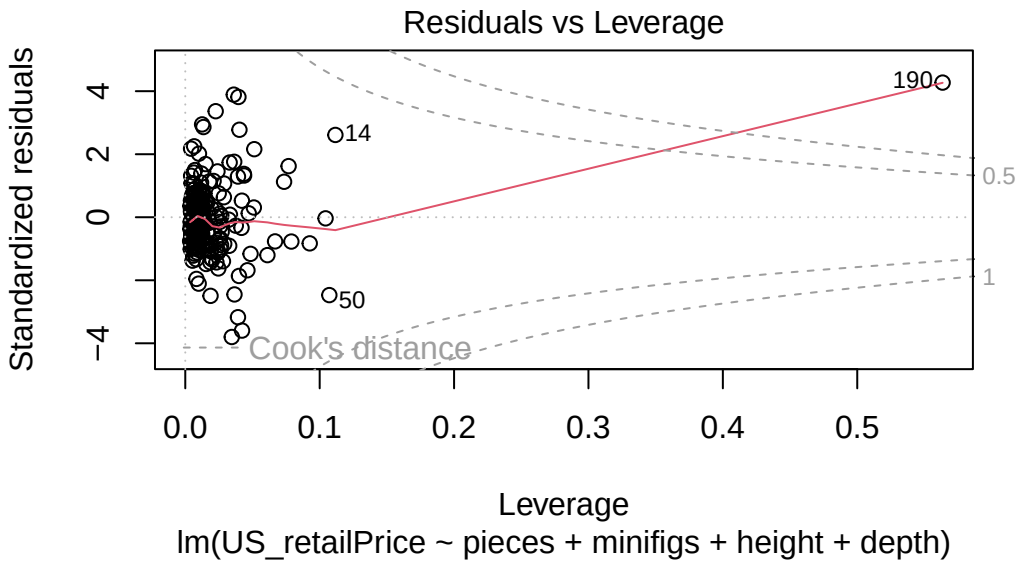
```r
# fit stage 1 model 4
model4 <- lm(US_retailPrice ~ pieces + minifigs + height + depth, data = df_stage1)

# plots
par(mfrow = c(1,1))
plot(model4)
```




Q−Q Residuals
Standardized residuals
−4
−2
0
2
4
−3
−2
−1
0
1
2
3
190
Theoretical Quantiles
lm(US_retailPrice ~ pieces + minifigs + height + depth)

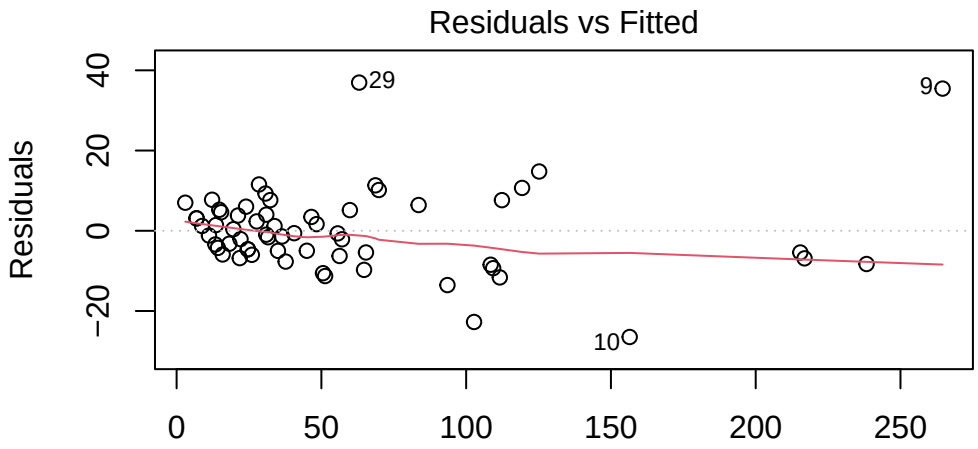

Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
0
200
400
600
800
190
Fitted values
lm(US_retailPrice ~ pieces + minifigs + height + depth)

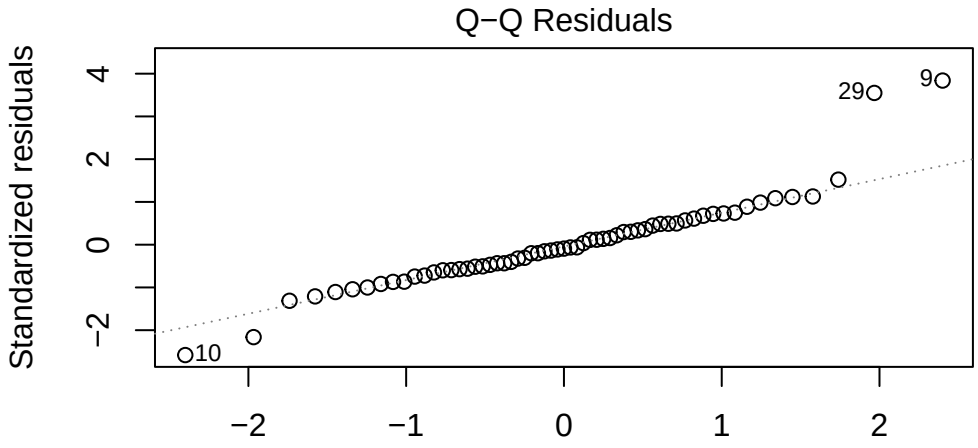


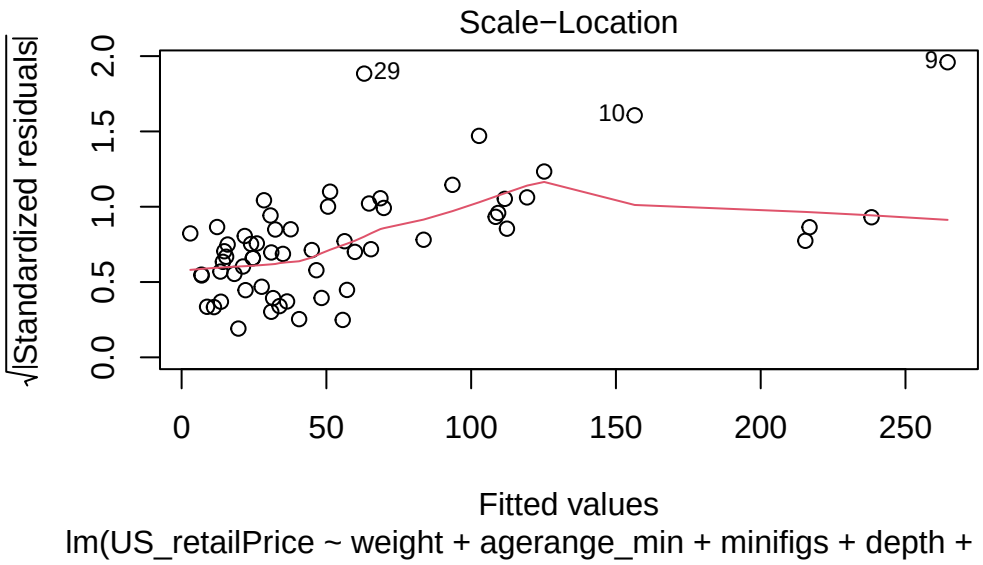
Stage 1 residuals vs fitted plot shows a cloud of points that are mostly homoskedastic, but the red LOESS line curves upward due an influential point. That point shifts the fitted values to the right which makes the residuals appear to fan out.

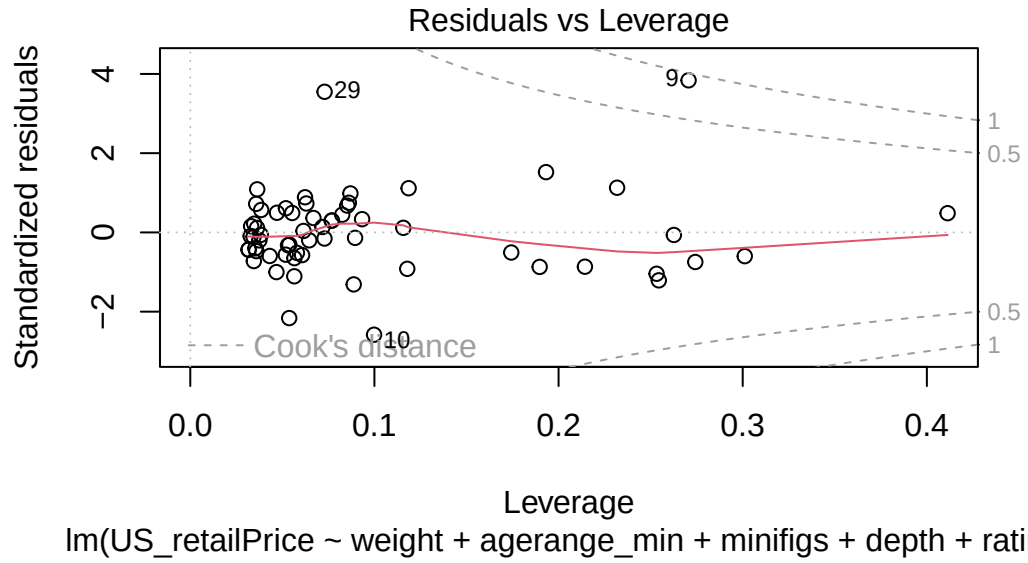
```
#fit stage 2 model 5
model5 <- lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating, data = df_st

#plots
par(mfrow = c(1,1))
plot(model5)
```


Residuals vs Fitted
Residuals
40
20
0
−20
29
9
10
0
50
100
150
200
250
Fitted values
lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating


Q−Q Residuals
Standardized residuals
4
2
0
−2
29
9
10
−2
−1
0
1
2
Theoretical Quantiles
lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating





The same influential point exists in Stage 2 diagnostics as it was in Stage 1, but the predictors, weight especially, explain price better so the point has less leverage. As a result, the residual patterns look more homoskedastic.

Diagnostic plots reveal meaningful differences in model behavior between the Stage 1 and Stage

2 models. In Stage 1, the residuals vs fitted plot displays some fanning, and the scale–location plot shows a mild upward trend. However, closer inspection indicates that this pattern is largely driven by a single influential observation, which pulls the line upward and shifts the fitted values. Aside from this point, the residual spread is relatively stable, and the apparent heteroskedasticity is more reflective of leverage effects.

In Stage 2, the residuals vs fitted and scale–location plots appear more homoskedastic, with a more uniform spread of residuals across fitted values. The same influential point is present, but the stronger predictors, reduces its leverage and diminishes its impact on the diagnostic plots. The Q–Q plot again follows the reference line closely. Despite the smaller sample size, the Stage 2 model exhibits cleaner diagnostic behavior, suggesting that the richer predictor set improves model stability.

These differences highlight the trade-off between sample size and predictor richness. Stage 1 benefits from broader coverage but is more sensitive to influential observations, while Stage 2 sacrifices coverage but gains a more stable residual structure.

8. Final Model Selection

The staged model-selection process produced two candidate models: a high-coverage Stage 1 model based on 289 observations and a richer Stage 2 model based on 61 complete cases. Stage 1 identified a four-predictor model including pieces, minifigs, height, and depth as the best model according to adjusted R^2, Cp, and BIC. Stage 2 identified a five-predictor model including weight, agerange_min, minifigs, depth, and rating as the best model, with the lowest Cp and BIC and an adjusted R^2 nearly identical to the larger six-predictor alternative.

To determine the final model, we compared diagnostic plots for both candidates. The Stage 1 model exhibited mild heteroskedasticity in the residuals vs fitted and scale–location plots, largely driven by a single influential observation that pulled the smoothing line upward. Aside from this point, the residual spread was reasonably stable, but variance increased slightly at higher fitted values. In contrast, the Stage 2 model displayed cleaner diagnostic behavior, with more uniform residual variance and reduced influence from the same high-leverage point. The inclusion of weight, the strongest predictor in Stage 2, appears to stabilize the residual structure and improve model fit despite the smaller sample size.

Although Stage 1 benefits from broader coverage, the Stage 2 model provides a more stable residual pattern and incorporates predictors with stronger explanatory value. The difference in adjusted R^2 between the Stage 2 five- and six-predictor models is negligible, and the five-predictor model is preferred based on Cp and BIC. Therefore, the final model selected for interpretation is the Stage 2 five-predictor model:

```r
#regression equation
summary(model5)
```

```
Call:
lm(formula = US_retailPrice ~ weight + agerange_min + minifigs +
    depth + rating, data = df_stage2)

Residuals:
    Min      1Q  Median      3Q     Max
-26.482  -5.907  -0.974   5.150  36.940

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.3865     7.4869  -0.052   0.9590
weight        58.8252     3.5662  16.495  < 2e-16 ***
agerange_min   2.9673     0.4384   6.768 9.04e-09 ***
minifigs       2.7778     1.1134   2.495   0.0156 *
depth         -2.8752     1.1516  -2.497   0.0156 *
rating        -1.8362     0.7603  -2.415   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.81 on 55 degrees of freedom
Multiple R-squared:  0.9695,    Adjusted R-squared:  0.9667
F-statistic: 349.7 on 5 and 55 DF,  p-value: < 2.2e-16
```

```r
# load package and check for collinearity
library(car)
```

```
Warning: package 'car' was built under R version 4.5.3

Loading required package: carData

Warning: package 'carData' was built under R version 4.5.3


Attaching package: 'car'
```

```
The following object is masked from 'package:dplyr':

    recode
```

```
vif(model5)
```

```
      weight agerange_min     minifigs        depth       rating
    5.608980     1.375048     2.448352     3.415140     1.177649
```

Final Regression Equation:

The intercept of -0.3865 represents the predicted price when all predictors are zero.Not meaningful substantively but is included for model completeness.

Weight (58.83) holding all other variables constant, each additional kilogram of weight is associated with an estimated $58.83 increase in retail price. Weight is the strongest predictor in the entire analysis ($p < 2e\text{-}16$). This aligns with intuition: heavier sets contain more material, larger builds, and higher production costs.

Minimum Age (2.97)for each one-year increase in the minimum recommended age, price increases by $2.97, holding other variables constant. Sets designed for older builders tend to be more complex, more detailed, and more expensive.

Minifigs (2.78) each additional minifigure increases price by $2.78, holding other variables constant. Minifigures are high-value components, often tied to licensed themes or collectible characters.

Depth (–2.88) for each additional centimeter of package depth, price decreases by $2.88, holding other variables constant.This negative coefficient is common in multivariate LEGO models as depth overlaps with weight and age range.

Rating (–1.84) for each one-unit increase in customer rating, price decreases by $1.84, holding other variables constant. This negative sign is expected: higher-rated sets tend to be better value, meaning they deliver more satisfaction per dollar. Rating reflects post-purchase sentiment, not production cost.

VIF values were all below 10, indicating no severe multicollinearity. Weight had a moderately elevated VIF (5.61), reflecting its overlap with other physical size variables, but this does not compromise model stability.

9. Limitations

Several limitations affect the interpretation of the final model. First, the Stage 2 model is based on only 61 complete cases, which reduces statistical power and may limit generalization. Sets with complete data tend to be larger, heavier, and more complex, introducing potential selection bias. Second, although diagnostics for the final model were generally favorable, one influential observation affected the residual structure in both Stage 1 and Stage 2, suggesting sensitivity to outliers.

10. Conclusion

This analysis used a staged model-selection strategy to evaluate predictors of LEGO retail price. Stage 1 used a larger, high-coverage dataset and identified physical characteristics such as piece count, minifigs, height, and depth as strong predictors. Stage 2 incorporated richer variables at the cost of reduced sample size. Diagnostics indicated that the Stage 2 model exhibited more stable residual behavior, and best-subsets criteria selected a five-predictor model including weight, agerange_min, minifigs, depth, and rating as the final model.

Weight emerged as the strongest predictor overall, reflecting its close relationship with set complexity and material cost. The final model balances explanatory strength and diagnostic stability, providing a an understanding of how physical attributes and customer sentiment relate to LEGO pricing. Despite limitations related to missingness and sample size, the staged approach offers a clear and interpretable analysis of the factors influencing retail price.